Deep Reinforcement Learning

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Lecture 1: Introduction and Course Overview

1.1 Sequential decisions

Reinforcement learning (RL) learns decision making from experience. An agent observes an environment, takes actions, and receives rewards; its actions influence future observations and opportunities. The aim is to learn behavior that achieves high reward over time.

Unlike standard supervised learning, RL generally does not provide the correct action for each input. Experience is also dependent on the agent's previous actions, rather than being a fixed collection of independent examples. Reward provides feedback about outcomes, but may not identify which earlier action was responsible.

Example: For a robot, observations can be camera images, actions can be motor torques, and reward can measure task success. For inventory management, observations include stock levels, actions are purchases, and reward can be profit. In both cases an action changes the situation in which the next decision is made.

1.2 Why deep reinforcement learning?

Deep reinforcement learning combines neural networks, which can learn representations of complex inputs and functions, with RL's framework for learning behavior. An end-to-end system can train components connecting sensory inputs to decisions against a shared task objective. A modular robot pipeline might instead separately estimate state, predict, plan, and control.

Examples include game playing and visual robotic manipulation. Deep models and RL do not automatically make real-world learning reliable: collecting experience, defining rewards, and transferring behavior remain difficult.

1.3 Beyond a specified reward

- Imitation learning learns to reproduce demonstrated behavior.
- Inverse reinforcement learning uses behavior to infer an objective that could explain it.
- Predictive models learn how the world changes and can help compare the consequences of candidate actions.
- Transfer learning reuses knowledge from other tasks; meta-learning aims to improve learning on new tasks.

1.4 Open questions

Deep RL can learn game strategies or sensor-based skills with sufficient experience, but often needs many interactions. Important questions include how to learn more efficiently, reuse knowledge, obtain suitable rewards, and use prediction effectively for control. Biological analogies offer motivation but do not establish that artificial agents and brains learn in the same way.

Source. Sergey Levine, Deep Reinforcement Learning, Fall 2021, Lecture 1.